

exactly 80 mm: _____

FIRST SOLVE

New here? Do the walkthrough once at cubepath.5z.vc/el/app/ learn. This card is the memory jog.

NOTATION

R U F L D B = turn that face clockwise, looking at it from outside the cube
 ' = counter-clockwise, 2 = half turn. Lowercase r f = that face plus the layer behind it, turning together.
 y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

1 WHITE CROSS



White center DOWN. Bring the four white edges to it. Each edge's side color must match the center it touches. No algorithm – work it out. Can't see it? Build the four white edges on the YELLOW face first, each above its matching center, then turn each one down 180 degrees.

2 WHITE CORNERS

Find a white corner on top. Turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
R'U'



white sticker FRONT
L'U'L



white sticker UP run the first one once to knock it out, then look again
R'U'

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to.

goes RIGHT

U **R'U'U'** y **L'U'L**



goes LEFT

U **L'U'L** y' **R'U'R'**

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ R'U' sexy move
■ R'U'U' Sune
■ R'FR' sledgehammer
gap = change grip

FIRST SOLVE

Newbie? Don't do the walkthrough once at cubepath-5s.vercel.app/learn. This one is the memory jog.

NOTATION

R = turn the face clockwise, looking at it from outside the cube
r = counter-clockwise, 2 = half turn. Lowercase **r** = that face plus the layer behind it, turning together
y = spin the whole cube left, white staying down. A whole-cube turn solves nothing.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm your hands run as one unit

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Find a white corner on top. Turn U until it sits above the empty slot it belongs in, then match a picture below and repeat until it drops in.



white sticker RIGHT
R'U'R



white sticker FRONT
L'U'L



white sticker UP run the first one once to knock it out, then look again
R'U'R

3 MIDDLE EDGES

Top edge with NO yellow: turn U until its front color matches the center under it. The arrow shows which slot it goes to



goes RIGHT
U' R'U'U' y L'U'L



goes LEFT
U' L'U'L y' R'U'R

Edge stuck in the wrong slot? Run either one to kick it out, then place it.

■ **R**U'R sexy move
 ■ **R**U'R Sune
 ■ **R**FRF' sledgehammer
 gap = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left to right
FRUR'U'F'



Hook hook pointing front-right
fFRUR'U'f'

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow; rest clockwise
RU'R'U RU2R'



Anti-Sune $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' U'RUR'



PI $\frac{2}{3}$ yellow; headlights left
fFRUR'U'f' FRUR'U'F'

OLL CORNERS - continued



Headlights $\frac{2}{3}$ yellow; headlights at the back
R2UR'U2 RD'R'U2R'



2x Headlights $\frac{2}{3}$ yellow; headlights both sides
RU'R'U RU'R' URU2R'



Chameleon $\frac{2}{3}$ yellow net to each other
fFRUR'U'f' FR'F'



Bowtie $\frac{2}{3}$ yellow diagonal
fR'FRUR'U' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn, also finishes the case. Machine-checked, there is the worst.

Top not all yellow and nothing matches? Run Sune and read again.

No yellow edge at the end? Run the first one, then read again.

■ **RU'R'U** sexy move
 ■ **RU'R'U** Sune
 ■ **R'FR'** slide change
 ■ **g** = change grip

TWO-LOOK

OLL CROSS - yellow edges



Line bar left to right
fRUR'U'f



Hook hook pointing front-right
fRUR'U'f

OLL CORNERS - yellow face



Sune $\frac{1}{2}$ yellow; rest clockwise
RUR'U RU2R'



Anti-Sune $\frac{1}{2}$ yellow; rest counter-clockwise
RU2R' U'RUR'



PI $\frac{3}{2}$ yellow; headlights left
fRUR'U'f fRUR'U'f

OLL CORNERS - continued



Headlights $\frac{3}{2}$ yellow; headlights at the back
R2UR'2U RD'R'U2R'



2x Headlights $\frac{3}{2}$ yellow; headlights both sides
RUR'U RU'R' URU2R'



Chameleon $\frac{3}{2}$ yellow next to each other
rUR'U'r' FRF'



Bowie $\frac{3}{2}$ yellow diagonal
fRUR'U'r' FR

FALLBACK

The badge on each row is your fallback: that many Sunes, with a U turn between, also finishes the case. Machine-checked, there is the worst.
Tap not all yellow and nothing matches? Run Sune and read again.

No yellow edge at all? Run the first one, then read again.

■ **RUR'U** sexy move ■ **RUR'U** Sune ■ **R'FRF** sledgehammer **gap** = change grip

ONE-LOOK PILL

ADJACENT CORNER SWAP

exactly one face shows headlights

• T L headlights: pairs B-left F-left
RUR'U' R'F R2UR'U' RUR'F'

• J a L solid: pairs B-right F-left R-front
x R2FRF' RU2 rUr U2

• Jb L solid: pairs B-left F-back R-back
RUR'F' RUR'U' R'F R2UR'U'

• Aa L headlights: pairs F-right B-front
x L2D2 L'U'L D2 L'U'L

• Ab L headlights: pairs B-right R-back
x' L2D2 LUL' D2 LU'L

• F L solid
R'U'F' RUR'U' R'F R2UR'U' RUR'UR'

One face shows headlights, two corners of that face match. Hold as shown, turn the top face to finish.

■ R U R' ■ U R U' ■ S U S' ■ F R F' ■ S L S' ■ S R S' ■ S U S' ■ S D S' ■ S B S'

ADJACENT CORNER SWAP

continued

• Ra L headlights: pairs F-left
RUR'U' RUR DR'UR'D' R'U2R'

• Rb L headlights: pairs B-left
R2F RURU' F'R U2R'U2R

• Gc L headlights: pairs F-right
R2UR'U'R' RU'R2 U'D R'UR'D'

• Gb L headlights: pairs B-back
R'U'R UD' R2UR'U'R' RU'R2D

• Gc L headlights: pairs B-right
R2UR'U'R RU'R2 UD' R'U'R'D

• Gd L headlights: pairs F-front
RUR' U'D R2UR'U'R' UR'UR2D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ONE-LOOK PLI

ADJACENT CORNER SWAP

exactly one face shows headlights



• T L headlights: pairs B-left F-left
RUR'U' R'F R2UR'UR' RUR'F'

• J a L solid: pairs B-right F-left R-front
x R2FRF' RUR'U' rUr U2

• J b L solid: pairs B-left F-back R-back
RUR'F' RUR'U' R'F R2UR'U'

• A a L solid: pairs F-right B-front
x LZD2 L'U'L D2 L'U'L

• A b L solid: pairs B-right R-back
x' LZD2 LUL' D2 LU'L

• F L solid
R'U'F' RUR'U' R'F R2UR'U'

One face shows headlights: two corners of that face match. Hold as shown, turn the top face to finish.

RUR'U' ascends, moves, **RUR'U'** descends, **R'FR'F'** slides/changes, **uau** = change ring

ADJACENT CORNER SWAP

continued



R a L headlights: pairs D-front U-right
RUR'U' RUR' FR UR'UR'D' R'UZR'



R b L headlights: pairs B-left R-front
R2F RUR'U' FR U2R'U2R



G a L headlights: pairs F-right R-back
R2UR'UR'U' RUR'U' D'U' R'UR'D'



G b L headlights: pairs R-back R-front
RUR'U' UD' R2UR'UR'U' R'UR'U2D



G c L headlights: pairs B-right R-back
R2UR'UR'U' RUR'U' R'UR'U2D



G d L headlights: pairs R-front R-back
RUR'U' UD' R2UR'UR'U' UR'UR'U2D

Learn in the printed order, three a week, about six weeks. Ja and Jb on the same day – one is the mirror of the other.

ANNEX – BIG CUBES AND THE MAP

BIG CUBES 4x4 and 5x5

Reduce: build the 6 centers → pair the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY – never wider it is.

last 2 edges

4x4 PLL parity

4x4 OLL parity – 1 edge pair flipped – half the time

Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 Lw U2 Rw U2 Rw' U2 Rw' U2 Rw'

5x5 edge parity – 1 edge pair flipped – half the time – one move differs, 5x5 has no PLL parity

Rw U2 x Rw U2 Rw U2 **3Rw** U2 Lw U2 Lw U2 Rw U2 Rw' U2 Rw' U2 Rw' U2 Rw'

NOTATION

R U L D B = turn that face clockwise, from outside the cube. ' = anticlockwise.

M = middle slice, turning the way I turn. x = the whole cube, turning the way I turn. i = twice. i twice = twice twice i = twice plus the layer behind it, turning together.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm that you hands runs as one unit - solve move - the R U2 solve move - the R U2 trigger - the R U2 solve move - the R U2 trigger - parity = a case a 3x3 cannot have - edge pairs = two edges glued into one, on a 4x4 or 5x5

RURU move **RURU** SU **RFRF** sledhammer **gap** = change grip

ANNEX – BIG CUBES AND THE MAP

3x3x3 QGO A A
 BIG CUBES 4x4 and 5x5
 Reduce: build the 6 centers → put the 12 edges → solve it as a 3x3. Rw = 2 layers wide. 3Rw = 3 layers. 2R = 2nd Layer ONLY → never widen it.
 last 2 edges
 4x4 PLL parity
 4x4 OLL parity = 1 edge pair flipped - half the time
 Rw U2 x Rw U2 Rw U2 Rw U2 U2 Lw U2 Rw U2 Rw U2 Rw U2 Rw U2
 5x5 edge parity = 1 edge pair flipped - half the time - *one more* difference, 5x5 has no PLL parity
 Rw U2 x Rw U2 Rw U2 **3Rw** U2 Lw U2 Rw U2 Rw U2 Rw U2 Rw U2
3Rw = the last layer, before OLL and PLL - both parity allowed

NOTATION

R U F L D = turn that face clockwise, from outside the cube. = 'a' turn
 r u f l d = turn that face clockwise. z = half turn
 M = middle slice, turning the way l turns. y turns x = the whole cube, turning the way R turns. y' = the opposite face. i f = that face plus the layer behind it, turning together.

WORDS

algorithm = a fixed sequence of turns - trigger = a short algorithm that you hands runs as one unit - *slice move* = the R U F L D trigger - *turning the way R turns* = the R F L D trigger - a case a 3x3 cannot have - edge pair = two edges glued into one, on a 4x4 or 5x5
 RURU' = *sexy*
 RURU' = *sexy*
 RURU' = *sexy*
 FRFP = *sledgehammer* - gap - change grip

slice out, run it, slice back - both cubes

2 edge pairs swapped - half the time

BEGINNER SHORTCUT - use it until you know. Sune

Twist a yellow corner four turns at a time, turning ONLY the top face

yellow faces RIGHT **D'R'D x2**

yellow faces FRONT **D'R'DR x2**

Print at 100% / Actual size — do NOT fit to page. Two-sided, flip on the LONG edge. The numeral on both faces is your check: cut ONE card first. This bar must measure exactly 80 mm:

exactly 80 mm: _____

4 YELLOW CROSS

yellow edges only – ignore the corners



Dot no yellow edge at all
FRUR'U'F'



Hook two at a right angle: hold them BACK and LEFT
FRUR'U'



Line hold the bar left-to-right
FRUR'U'

One algorithm, up to three times: dot to hook to line to cross.

5 MATCH THE YELLOW EDGES

First turn it so at least two top edges match the center under them. This algorithm is called *Sune*; you use it on every card after this one.



two matched, side by side hold them BACK and LEFT
FRUR'U' R2U'



two matched, opposite run it once from any hold, then look again
FRUR'U' R2U'

6 PUT THE YELLOW CORNERS HOME



one corner home hold it FRONT-LEFT
RU'L U' R'L



no corner home run it once from any hold, then look again
RU'L U' R'L

This cycles the other three corners into place. It also twists the yellow face you just look expected, the next step rebuilds it.

7 TWIST THE YELLOW CORNERS



yellow faces RIGHT
R'D'R D' R2



yellow faces FRONT
D'R'DR x2

Between corners turn **ONLY** the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look distorted while you do this. Keep going – the last top turn brings it back.

LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently – it is the only card on this card that gets reset.

UNDOCK CARD 2 – five scramble in a row with this card face down. **0 MASTER:** those five under 300. My target: _____ NEXT: 2/3 TWO-LOCK

4 YELLOW CROSS

yellow edges only – ignore the corners



Dot no yellow edge at all
FRUR'U'



Hook two at a right angle: hold them BACK and LEFT
FRUR'U'



Line hold the bar left-to-right
FRUR'U'

One algorithm, up to three times: dot to hook to line to cross.

5 MATCH THE YELLOW EDGES

First turn U so at least two top edges match the center under them. This algorithm is called **Sune**; you use it on every card after this one.



two matched, side by side hold them BACK and LEFT
FRUR'U' R2U'

look again



two matched, opposite run it once from any hold, then
FRUR'U' R2U'

6 PUT THE YELLOW CORNERS HOME



one corner home hold a FRONT-LEFT
RU'L'U' R'L'



no corner home run it once from any hold, then look again
RU'L'U' R'L'

This cycles the other three corners into place. It also twists the yellow face you just moved. Expected, the next step rebuilds it.

7 TWIST THE YELLOW CORNERS



yellow faces RIGHT
R'D'R'D x2



yellow faces FRONT
R'D'R'D x2

Between corners turn **ONLY** the top face, to bring the next unsolved corner to front-right. Never turn the whole cube. The cube will look distorted while you do this. Keep going – the last top turn brings it back.

LAST-LAYER ORDER ON THIS CARD: yellow cross, match edges, place corners, twist corners. Card 2 replaces this order permanently – it is the only one this card that gets revised.

UNLOCK CARD 2 – five scrambles in a row with this card face down. 3: MASTER: those five under 300. My target: _____ NEXT: 2/3 TWO-LOCK

P1L CORNERS = headlits on 2 matching corners on one face

 T headlits on ONCE face – hold that face LEFT
RUR'U' R' R2UR'U' RUR'
 T-Perm. No headlits to hold left? Run it twice.

 Y no headlits – on the two swapping corners sit naturally
FRU'R' RUR' RUR' R'FR'
 F-Perm. No headlits to hold right? Run it twice.

P1L EDGES = corners home, top still not solved

 UZ the front edge travels LEFT
RU R'UR' R'U' UR'
 U-Perm. No headlits to hold right? Run it twice.

 Ua the front edge travels RIGHT
M2U M2U M2UM2
 M-Perm. No headlits to hold right? Run it twice.

 H both pairs swap straight across
M2UM2 U2 M2UM2
 M-Perm. No headlits to hold right? Run it twice.

 Z each pair swaps with its neighbour
M'U' M2UM2M'U' M'U2U M2U
 M-Perm. No headlits to hold right? Run it twice.

START HERE = two new algorithms finish any last layer
 Show corner F R U R' U' until the cross appears – at most 3.
 Yellow face: Same, read again, repeat – at most 3 (see the badges).
 Corners: T-Perm. No headlits to hold left? Run it twice.
 Edges: UO. No solved edge to put at the back? Run it twice.
 Each case on the front replaces one repeat with one run. Three a week you can solve the whole four faces.

HVSZ
 Both show headlits on all four faces. The edge between them decides if belongs to the opposite face = H, to a neighbour = Z.

CORNERS
 Stick home but nothing solves? Turn the top face once and read again – that is often the whole lot.
 One case has beaten you five times! Park it, use its fallback above, come back next week.
 A single piece locks twisted and no algorithm touches it? The cube was reassembled wrong. Pop it out and restart it.

WORDS
 OLL = make the whole top face yellow - PLL = slide the top pieces home - headlits = two matching corners on one face - AUF = the free top turn that lines a case up

PLL CORNERS: headlights = 2 matching corners on one face

 T headlights on ONE face – hold that face LEFT
RUR'U' R'F R2UR'U' RUR'F'

 Y no headlights – the two swapping corners sit naturally
FRU'R' RUR'F' RUR'U' R'FR'

PLL EDGES: corners home, top still not solved

 Uh the front edge travels LEFT
R2U RUR'U' R'U' RUR'

 Uu the front edge travels RIGHT
M2U MU2 MU2M

 If both pairs swap straight across
H2UHU2 M2U MU2M

 Z each pair swaps with its neighbour
M'U' MU2UR2U' M'U2 U2U

THE NEW ORDER: yellow face, yellow face, corners home, edges home. This never changes again. Card 1's order is retired, and so is Niklas: it moves corners but wrecks the yellow face, and here the yellow face is finished first.

UNUSUAL CARD 3 – different last layers in a row, each finished in exactly two algorithms, case named out load first. A repeat does not count. **MASTER:** solves averaging under 1:00. My target: **~1:00** – NEXT: 3/3 ONE-LOOK PLL

EDGES ONLY: corners already home

- **Ua** B solid: F.L.R headlights: 3 edges counter-clockwise
M2U MU2 M'UM2
- **Ub** B solid: F.L.R headlights: 3 edges clockwise
M2U RUR'U' R'U' R'UR'
- **H** all 4 headlights: opposite edges swap
M2U'M2 U2 M2U'M2
- **Z** all 4 headlights: adjacent edges swap
M2U' M2UM2U' MU2 M2U

DIAGONAL CORNER SWAP – no headlights anywhere

- **Y** pairs F-left R-back
FRU'R'U' RUR'F' RUR'U' R'RF'
- **Y** pairs F-left L-front
Y' pairs U' R'F' R2UR'U R'RF'
- **No** pairs R-back F-right L-back R-back
RUR'U' RUR'F' RUR'U' R'F
R2UR'U' UZUR'U'
- **Nb** pairs R-right F-left L-back R-front
R' URUR' F'U'F RUR' FR'F'
R'UR'
- **X'** ending matches
E' L'U'D' L'U'D' L'U'D' L'U'D

HOW TO LOOK

- Count the headlight faces. One = front of this card. Four = this column. None = the column on the right.
- Only then read the edges to pick the exact case.
- Line the case up as drawn, rank it, turn the top again to finish – that last free turn is the AUF.
- Always yours from Card 2.

WORDS

PLL = slide the top pieces home • headlights = two matching corners on top • AUF = one free top turn that makes a case up • adjacent corner swap = the swapping corners share an edge • diagonal corner swap = the swapping corners are opposite

Card 2 gives you 19/72 last layers in one look. This card gives 7/72. Next: the other 7 top-faced cases, and the first two twenty-one rows as pairs – drilled with a cube in the app: cubeplus3x3x3vercel.app/practice

Done! All twenty-one named correctly. Twenty-ones in a solved, no pairs band, on two separate days, on two separate days. All solved under 0:30. My target: 0:20.

EDGES ONLY - corners already home

- **Ua B solid; F L R swap; 3 edges counter-clockwise**
M2U' MU2 M'UM2
- **Ub B solid; F L R headlights; 3 edges clockwise**
R2U R'UR' R'U' R'UR'
- **H all 4 headlights; opposite edges swap**
M2U'M2 U2 M2UM'2
- **Z all 4 headlights; adjacent edges swap**
M'U' M2UM2U' MU2 M2U

DIAGONAL CORNER SWAP - no headlights anywhere

- **Y pairs F-left R-back**
FRU'R'U' RUR'F' RUR'U' R'RF'
- **V pairs F-left L-front**
R'UR'U' y R'F R2UR'U R'RF'
- **Na pairs R-right L-back R-back**
RUR'U' R'UR'F' RUR'U' R'F
R2U'R' U2R'U'R'
- **Nb pairs B-right L-left L-back B-front**
R'U URU'R' F'UF' RUR'F' R'F'
R'UR'
- **E nothing matches**
X 'L'UD' L'U'LD L'U'LD' L'UD'

HOW TO LOOK

- Count the **headlight** faces. One = front of this card. Four = this column. None = the column on the right.
- Only then read the edges to pick the exact case.
- Line the case up as drawn, run it, turn the top again to finish – that last free turn is the AUF.

• already yours from Card 2.

WORDS

PLL = slide the top pieces home - headlights = 2 matching corners on one face; AUF = 180° free turn that makes the case up-adjacent corner swap = the swapping corners share an edge - diagonal corner swap = the swapping corners sit opposite

Card 2 gives you 19/22 last layers in one look. This card gives 71/72. The other 51 off 72-face cases, and the first two layers solved on pairs – drilled with a cube in the app: cubeapp315s.vrered.app practice under DONE – all twenty-two named correctly, twenty-one in a row, no pairs in row, on two separate days, on two separate days at 0.30, my target.

THE LADDER

1/3 FIRST LOOK - 5 algorithms - unlock: five solves, card face
 2/3 TWO-LOOK - 11+ unlock: fifteen last letters in exactly two
 3/3 ONE-LOOK PL1 - +15: done! all 21 named, twice on separate
 days

So that: full OLL, F2L, cross planning, look-ahead – in the app
 not on card stock.

WHY THERE ARE ONLY THREE

1. The first two are the only ones you tell apart by sight. It is
 bad at this you drill F2L, cross planning and look-ahead are drills
 with no finite case tell. Full OLL is 57 cases, 22 of which differ only
 by a diver 4 fifth of a millimetre wide at this size. All of them are
 in the app, with a real cube and randomised setup. This set stops
 where the medium stops.

PRINT AND VERSION

EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were
 to wonder the build would follow the same path. Recognition wording is generated from the same permutation diagram is drawn from.

Set V1.0: **reprint** any single card at cubepu3h-s1s.vercelapp/print - print at 100% / Actual size, never Fit to page!

Not a tier, no unlock, no record. Use it when you buy a 4.

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THE LADDER
13 FIRST Solve - 9 algorithms - unlock: five solves, card face
down - 100% - Actual size, never Fit to page.
20 TWO-LOCK - 135 - unlock: fifteen last layers in exactly two
moves - 100% - Actual size, never Fit to page.
35 THREE-LOCK PLUS - 115 - done: all 21 named, twice on separate
days - 100% - Actual size, never Fit to page.
40 FOUR-LOCK PLUS - 135 - done: all 21 named, twice on separate
days - 100% - Actual size, never Fit to page.
WHY THERE ARE ONLY THREE
There are three kinds of cases you tell apart by sight. It is
bad at this you drill. Full OLL, cross planning and look-ahead are drills
with no finite case table. Full OLL is 57 cases, 22 of which differ only
in a driver a fifth of a millimetre wide at this size. Full PLL is 42
cases, 22 of which differ only in a driver a fifth of a millimetre
wide at this size. Full PLL is 42 cases, 22 of which differ only
in the app, with a real cube and randomised setup. This set stops
where the medium stops.
PRINT AND VERSION
EVERY ALGORITHM on these cards is expanded from machine-verified data and checked against a cube simulator at build time; if one were
willing to build would be a good idea. Recognition wording is generated from the same pronunciation diagram is drawn from.
Set v1.0 - reprint any single card at cubephub-3d-vercelapp/print - print at 100% / Actual size, never Fit to page.
Not a tier, no unlock, no order. Use it when you buy a 4x4.

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Print at 100% / Actual size – do NOT fit to page. Two-sided, flip on the LONG edge. The number of pages is 10. The width of the page is exactly 80 mm: _____